

ARNOB RIZWAN AHMAD

AI / LLM Engineer • Agentic Systems • RAG & Evaluation • Python Backend • Cloud
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PROFILE

AI/LLM engineer and Software Engineering graduate of Universiti Teknologi Malaysia (July 2026). Build production LLM systems end to end — agentic workflows, RAG over FAISS and pgvector, and evaluation harnesses with blinded LLM-judge scoring and execution guardrails — plus the FastAPI, PostgreSQL and Kubernetes infrastructure they run on. Seeking AI/ML engineering roles where evaluation and reliability matter as much as model choice.

EDUCATION

Universiti Teknologi Malaysia (UTM) — Johor Bahru Oct 2022 — July 2026
Bachelor of Software Engineering (Computer Science) — Graduated July 2026

PROFESSIONAL EXPERIENCE

Agentic AI Engineer — BLOOMTECH, Inc. — Remote (Part-time) Aug 2026 — Present
Agentic AI systems and LLM-driven tooling

- Build agentic AI workflows in Python — multi-step LLM planning, tool and API integration, and structured output handling.
- Design prompt and evaluation loops that measure and improve agent reliability across task categories.

Enterprise Resource Planning Developer — Hirdaramani Bangladesh — Hybrid (Full-time) Jul 2026 — Present
ERP systems for a multinational apparel manufacturing group

- Build and maintain ERP modules supporting production, inventory and order-management workflows across factory operations.
- Write and tune SQL against large operational datasets; build reports and data integrations between ERP and internal systems.
- Work with operations and IT stakeholders to translate process requirements into shipped features.

Application Developer (Freelance) — Moien Delivery — Luxembourg (Remote) Mar 2026 — Jul 2026
Hybrid logistics platform serving customers, restaurants and drivers — live on Google Play & Apple App Store

- Built the Angular web front-end and a cross-platform React Native mobile app on a Nuxt.js backend, delivering features for three distinct user roles.
- Implemented real-time order tracking, driver dispatch and delivery workflow automation over REST APIs and WebSocket event streams.
- Owned features from spec through QA to production deployment on weekly release cycles with international stakeholders.

Machine Learning Engineer (Contract via Mercor) — AI News HUD — USA (Remote) Oct 2025 — Mar 2026
US-based AI-powered news intelligence platform

- Built and evaluated production PyTorch models for real-time news classification, summarization and personalized recommendations.
- Designed evaluation pipelines and ran iterative experiments on live traffic, raising model accuracy to **78.9%** and cutting inference latency by **20%**.
- Took ML components from prototype to deployed feature in the platform's core intelligence layer.

Freelance & Contract Engineering Sep 2024 — Jul 2026
Concurrent client and programme work delivered alongside study

- **Framic** (remote, Aug 2025 — Jul 2026) — founding backend engineer at an AI storage startup; designed backend services from scratch, owned architecture and core API design, and led AI search and real-time file indexing.
- **MOMS Mobile Oil Change** (USA, Apr — Jul 2026) — production Next.js site, live at momsoilchange.com: rebuilt navigation and three product pages, led an SEO overhaul, and ran a site-wide security and dependency audit.
- **Headstarter** (USA, Sep 2024 — Mar 2025) — competitive 6-month software engineering residency; shipped 14+ ML and full-stack projects, trained 5 neural networks, and cut image size and build time on a Docker-based GitHub Actions pipeline.

SELECTED PROJECTS

School Care App — 40-School Deployment — Sylhet, Bangladesh Mar 2026
• Delivered a full-stack school management system adopted by 40 schools across Sylhet; in closed testing ahead of Google Play launch.

- Built a React admin dashboard and a React Native app for parents and teachers, with Cloud Functions fanning out role-targeted FCM push notifications by school, role, and class.

ShopLocalr — Local Commerce Mobile App — Malaysia Feb 2026
• Shipped a 33,000-line Flutter local-business discovery app with customer, business, and admin roles — QR deal redemption with checkout PINs, Gemini-backed recommendations, and offline-first Hive caching on a Firebase backend.

Autonomous ML Agent — LLM-Planned AutoML Pipeline (GitHub)
• Built an AutoML system where an LLM planner proposes modeling strategies under a time budget, Optuna tunes XGBoost/LightGBM/scikit-learn models, ensembles are built, and SHAP explains the winner.
• Shipped with model cards, a FastAPI inference service, 5 pytest suites, and CI/CD publishing to Docker Hub.

Self-Improving Prompt Optimization API — LLM Evals Infrastructure (GitHub)
• Built CI/CD for prompts: a semver-versioned prompt registry where each improvement cycle runs failure analysis, generates candidates, A/B-evaluates them with a blinded LLM judge, and auto-promotes winners per YAML rules.
• Multi-dimensional scoring across OpenAI and Anthropic providers with FastAPI, Postgres, and Redis.

CrisisCohort — Multi-Agent AI Climate Resilience Platform (GitHub)
• Architected a climate-resilience platform for smallholder farmers with 10+ specialized AWS Bedrock agents (water optimization, climate migration, grant matching, disease diagnostics) across 15,000+ lines of Lambda.
• Designed an AWS architecture spanning Lambda, Bedrock, DynamoDB, CloudFront, API Gateway, and Cognito, plus an on-chain water-rights trading contract in Solidity.

Terminal Coding Agent — Sandboxed CLI Agent (GitHub)

- Built a pip-installable CLI coding agent that plans edits via OpenAI/Anthropic, applies them through a diff manager, and executes inside E2B Firecracker microVMs.
- Added guardrails scanning for six secret types (AWS, GitHub, Slack keys) and 5 pytest suites.

TECHNICAL SKILLS

ML / AI: PyTorch, scikit-learn, XGBoost, LightGBM, Optuna, SHAP; LLM agents, RAG (FAISS, pgvector), LLM evaluation & blinded judges, prompt engineering, AWS Bedrock, Gemini

GPU / HPC: AMD ROCm, HIP, CUDA→HIP porting, rocprofv3 profiling, mixed-precision (FP16/BF16) GEMM, kernel tuning

Languages: Python, SQL, TypeScript, JavaScript, C#, Java, Dart

Backend: FastAPI, Django + DRF, Node.js, Express, NestJS, ASP.NET Core, Celery, REST APIs, WebSockets

Frontend & Mobile: React, Next.js, Angular, React Native, Flutter

Cloud & Data: AWS, Microsoft Azure, GCP, Docker, Kubernetes, Linux, CI/CD (GitHub Actions, Jenkins); PostgreSQL, MySQL, SQL Server, MongoDB, Redis, ERP modules and SQL tuning on large operational datasets

CERTIFICATIONS

- **AMD — ROCm Certified Associate** (Aug 2026)
- **AWS** — Building Language Models on AWS; Customizing & Evaluating LLMs Using Amazon SageMaker JumpStart; Amazon SageMaker JumpStart Foundations; Foundations of Prompt Engineering (Mar 2025)
- **IBM** — Full Stack Software Developer Professional Certificate (Aug 2024)
- **Microsoft Azure AI** — Azure OpenAI Service, Natural Language Processing, Document Intelligence (Jun 2024)